

But now compare this to the case of the multiplicative

Which of these, if either, does the multiverse hypothesis make more likely?
Only E2, it seems. But it looks like E1 is the evidence we have; so it looks like the probability of our evidence conditional on the multiverse hypothesis is not higher than the probability of that evidence conditional on the single universe hypothesis.

E1. This
universe has
life-
permitting
constants.

E2. Some
universe has
life-
permitting
constants.



fine-
tuning of the
universe

A solid red circle with a subtle drop shadow, centered on a white background. Inside the circle, the text "Bayes' theorem" is written in a white, sans-serif font, centered both horizontally and vertically.

Bayes'
theorem



the
argument



objections

Could one object that we also have E2 as evidence, and say that E2 is evidence for the multiverse?

But if that were legitimate we could do the same thing in the many rollers
case -- and we know that that is a mistake.

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Which of these, if either, does the null hypothesis make more likely?

universe hypothesis.

likelihood probability of our evidence conditional on the null hypothesis is

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contemporary physics which will be important to the argument which

Enough about probability (for now). Let's turn now to some results from

follows.

A first question: what does the theory provide by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, “Fine-tuning fine-tuning”)



These 'dimensional constants' will be our focus.

One such constant is the cosmological constant, which measures the

energy density of empty space.

The cosmological constant is just a number which (as far as we know)

could have different values consistent with the laws of physics.

One thing that the standard model of physics gives us is a measure of how

(like the cosmological constant) would fall in a certain range.

likely it is, given the laws of the nature, that the fundamental constants

We can make certain plausible assumptions about what it would take for

hard to see how life could have evolved. If there were no planets, it is hard

life to have evolved. For example, if there were nothing but hydrogen, it is

to see how life could have evolved.

Given assumptions such as these, we can look at what the standard model

constant has a value which would permit the evolution of life.

of physics tells us about how likely it is that, for example, the cosmological

“Physicists have determined the (approximate) values of the fundamental constants by measurement.

(There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

supporting value, given the laws of nature, is very low.

This is the claim that the cosmological constant having a life-

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. “Small” here doesn’t mean “1 in 10,000” or “1 in 1,000,000”. It means the kind of fraction that one would resort to exponents to describe, as in “1 in 10 to the 120”. The kind of package that we have in mind tells us that only a fantastically small range is life permitting.



nature, is

cosmological constant falling in a life-supporting range, giving the laws of

The claim is that, according to current physics, the probability of the

1



10120

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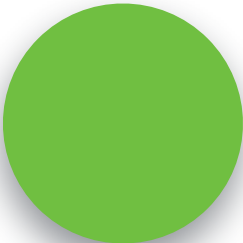


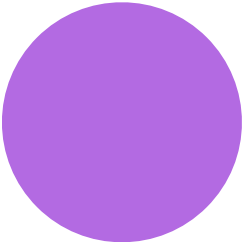
fine-
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Bayes'
theorem







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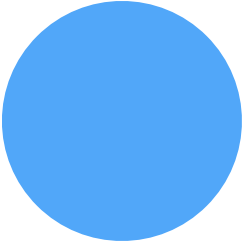
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the
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The claim is that, according to current physics, the probability of the

completing a constant falling in a life-supporting range, given the laws of

1

10120

probability theory into an argument for the existence of God.

Let's now ask how to turn these facts about current physics and

of the universe.

Much as we considered two different hypotheses about the number of balls

in the turn, so we can consider two different hypotheses about the origins

Let the design hypothesis be the hypothesis that the universe

hypothesis that it was not.

created by an intelligent designer. Let the non-design hypothesis be the

Let e be the fact that the cosmological constant has a

value in the life-permitting range.

Then what do we know about relevant probabilities?



$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$



$$Pr(e | \text{design}) \equiv 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

design hypothesis given evidence is that just plugging in the

numbers.

With this information in hand, figuring out the probability of the non-



$$Pr(\text{non-design}) = 0.5$$



$$Pr(\text{design}) = 0.5$$



$$Pr(e) = 0.25$$



$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

It is difficult to think about numbers as large as the denominator of this fraction. But

to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now

PowerBar!



consider the odds of winning Powerball one trillion times in a row. Call that a “super

Now consider the odds of winning a super Powerball one trillion times in a row. Call

that a 'super duper Powerball.'

Now consider the odds of winning a super Powerball one trillion times in a row.

The odds of this happening are about $1/10^{44}$ — so much, much higher than the

odd of the universe being life-permitting by chance.

This means that if you simply take the physics at face value, and begin by assigning a

superduperpowerball a trillion times in a row.

probability of 0.5 to the non-design hypothesis, you should think that the chances of

the non-design hypothesis being true are vastly lower than the chances of winning a

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1.

minus the very small number we're just discussing:

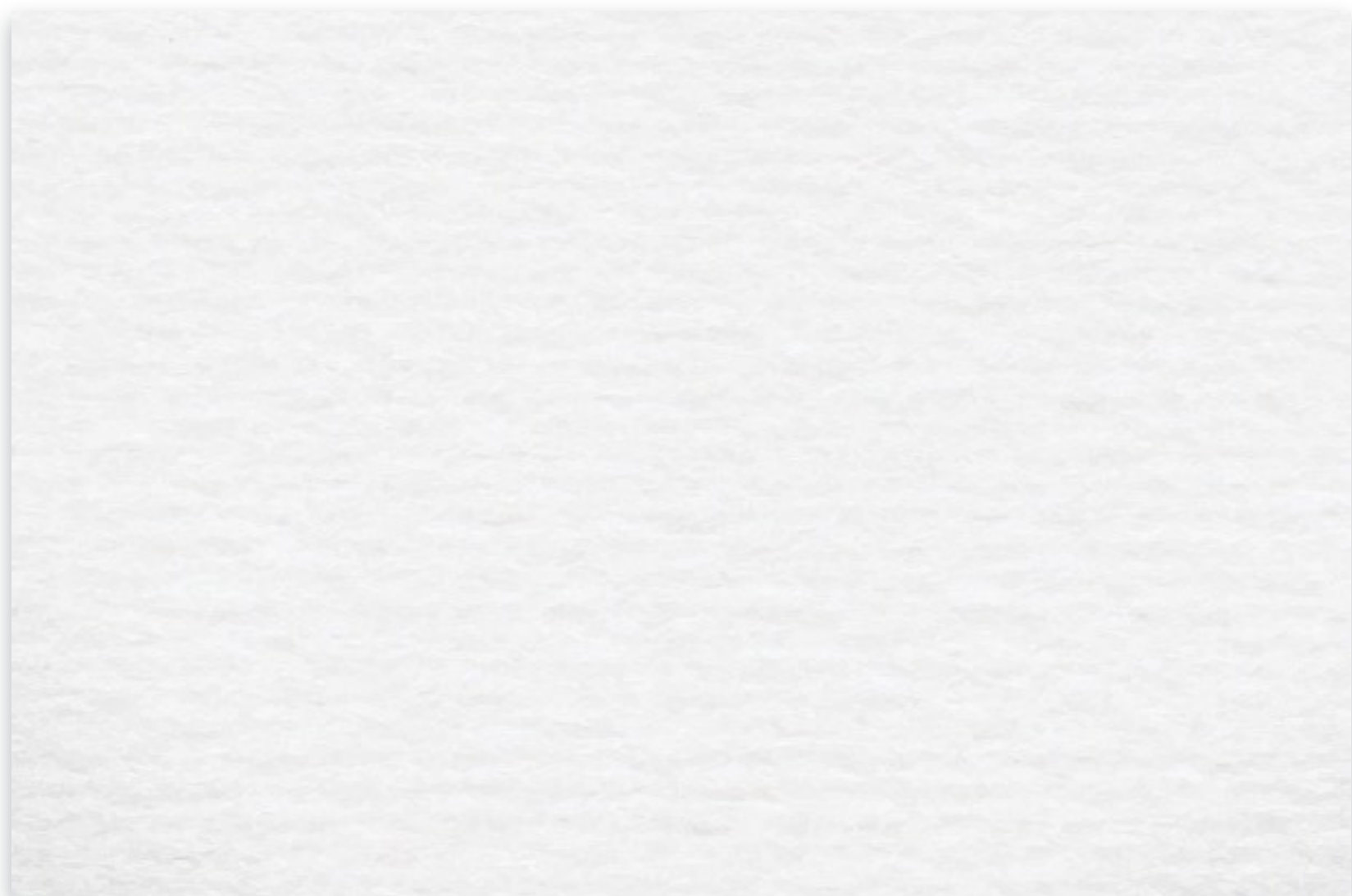


$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This is about as close to certainty as it is possible to get.



objections



2



P



$$3. P(\text{design}) = P(\text{non-design}) = 0.5.$$

4. Bayes', theorem.

5



P

6. *Life exists.*

8. If the universe was created by an

$$7. P(\text{design}) \approx 1. \quad (5, 6)$$

$$\text{C.P.}(\text{God exists}) \approx 1. (7,8)$$

1, 2, 3, 4

independent designer, then God exists.

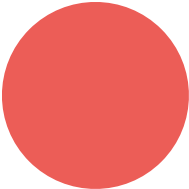


$$(life/deSIGN) \equiv 0.5$$

$$(1\text{ife} \mid \text{non-design}) = \frac{1}{10^{120}}$$

$$(\text{design} / \text{life}) \approx 1$$







fine-
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